

3

Customer acquisition

3.1 Introduction

As many companies have transited from brand-centric to customer-centric, the management of customer relationships has become especially important. Much research attention has also transferred from the product life cycle to the customer life cycle. Along the customer life cycle, the essential processes of CRM can be separated into four circulatory steps: customer acquisition, customer retention, customer churn, and customer win-back. Customer acquisition, the first step, is the foundation of the whole CRM process and is also a cornerstone in the development of the business of a company. The statement that acquiring a new customer is several times more costly than retaining an existing customer has clearly shown the importance of a successful customer acquisition program. And, without sustained customer acquisition there will eventually be no customers left for the company to try and retain. Despite the significance of customer acquisition, research on this topic is still not sufficient. This chapter seeks to review and summarize the limited research on customer acquisition and provide guidelines on how a firm can use its own prospect and customer data to aid in successful and profitable customer acquisition.

In the past 15 years, researchers have paid attention to the key issues which occur in the customer acquisition process. Questions that have been considered, as shown in Figure 3.1, include:

- How likely is it that the prospects will respond to our acquisition promotion?
- How many new customers can we acquire in this campaign?
- How many orders will each of our newly acquired customers place?

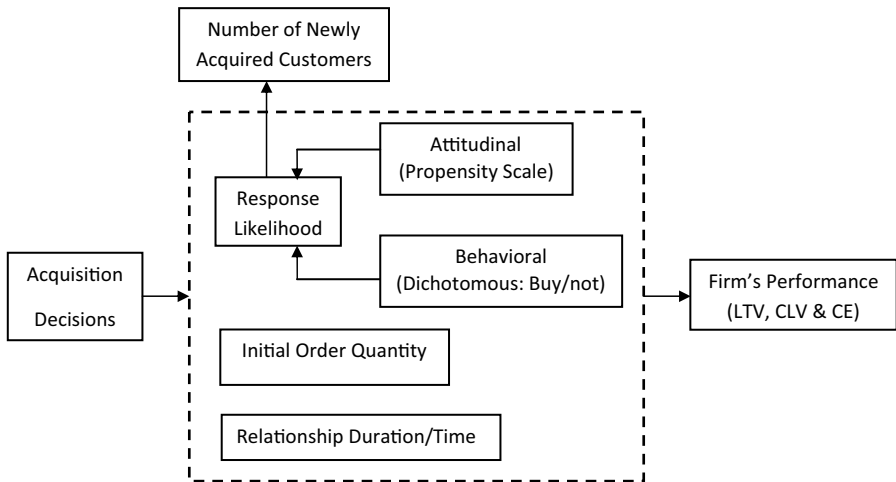


Figure 3.1 Issues addressed in customer acquisition models (LTV, Lifetime Value; CE, Customer Equity).

- How do the marketing variables, such as shipping fee, word-of-mouth referral, and promotion depth, influence prospects' response behavior?
- How long will the newly acquired customers stay with our companies?
- How much profit or value will this acquisition campaign bring to our companies?

Many studies have been conducted and many models have been developed to try and solve these questions. Table 3.1 lists a representative set of studies that have considered these issues and accounted for many of the problems that might occur in the model-building process. To provide a comprehensive understanding of how to model customer acquisition, we will review the issues in the studies one by one along with the related modeling techniques (Table 3.2). We will also provide empirical examples at the end of each subsection to demonstrate how to apply this knowledge to a representative sample of customers from a B2B firm.

Due to the differences in how the data can be collected and modeled, the first question that often needs to be answered is whether the company has customers that purchase in a non-contractual or contractual manner. In many instances this will determine the type of statistical model that needs to be used in order to gain any insights from the data.

In non-contractual business settings, such as for retailing firms and catalog companies, customers can freely split their spending across several firms and make purchases from these firms either occasionally/irregularly or frequently/regularly. The key difference from a modeling perspective is that in all non-contractual situations, the firm never actually observes when a customer is no longer a customer. Thus, once the customers make the first purchase from the company, it is uncertain how

Table 3.1 Summary of select empirical studies modeling customer acquisition.

Studies	Probability/ no./time	LTV/ CLV/CE	Conceptual/ empirical	Selection bias correction	Endogeneity correction	Censoring accounted	Explanatory variables	Industry type
Wangenheim and Bayón (2007)	Yes	Yes	Empirical	No	No	No	WOM and moderators	Energy market
Lix <i>et al.</i> [1]	Yes	No	Empirical	No	No	No	Demographics and life-style variables	Retailer
Lewis [2]	Yes	No	Empirical	No	Yes	No	Shipping, pricing and WOM	Online retailer
Villanueva <i>et al.</i> [3]	Yes	Yes	Empirical	No	Yes	No	Acquisition channels and their interactions	Internet log-in
Lewis [4]	Yes	Yes	Empirical	No	No	Yes	Discount depth and RFM	Online retailer
Hansotia and Wang [5]	Yes	Yes	Empirical	No	No	Yes	Historical transactions	Motor club membership
Thomas [6]	Yes	No	Empirical	No	No	Yes	Historical transactions	Membership
Reinartz <i>et al.</i> [7]	Yes	Yes	Empirical	Yes	No	Yes	Customer actions	High-tech manufacturer

	Yes	Yes	Empirical	Yes	No	Yes	Customer actions and characteristics of firm	B2B firm, pharma, and catalog retailer
Thomas <i>et al.</i> [8]							—	—
Blattberg and Deighton [9]	No	Yes	Conceptual	No	No	No	—	—
Berger and Bechwati [10]	No	Yes	Conceptual	No	No	No	—	—

WOM, Word Of Mouth; RFM, Recency, Frequency and Monetary value.

Table 3.2 Review of customer acquisition models.

Research interest	Specification	Estimation	Representative studies
Probability of being acquired	Logit	MLE	Hansotia and Wang [5]
			Wangenheim and Bayón (2007)
	Probit	MLE	Reinartz, Thomas, and Kumar (2005)
	Linear Regression	OLS	Lix, Berger, and Magliozzi (1995)
No. of newly acquired customers	Log-linear	MLE	Lix, Berger, and Magliozzi (1995)
	Systems of linear regression	3SLS	Lewis [2]
	Vector autoregression		Villanueva, Yoo, and Hanssens (2008)
Initial order quantity	Systems of linear regression	3SLS	Lewis [2]
Duration/time	Hazard function		Lewis [4] Schweidel, Fader, and Bradlow (2008)
Firm's performance (LTV, CLV, and CE)	Tobit	MLE	Lewis (2006)
			Hansotia and Wang [5]
			Reinartz, Thomas, and Kumar (2005)
			Thomas, Reinartz, and Kumar (2004)
	Vector autoregression		Villanueva, Yoo, and Hanssens (2008)
	Decision calculus	Deterministic	Blattberg and Deighton [9]
	Decision calculus	Deterministic	Berger and Bechwati [10]

MLE, Maximum Likelihood Estimation; OLS, Ordinary Least Squares; 3SLS, Three-Stage Least Squares.

often they will buy again and for how long they will repeat-purchase from that company. Thus, companies in non-contractual business settings need to consider the probability that new customers will be attracted to buy the first time, how many orders they will bring to the companies over time, and how much profit they will generate over their lifetimes.

In contractual business settings, such as newspaper subscriptions and telecommunications services, customers usually pay their subscription or service fees monthly or yearly and companies know when customers leave the relationship. Once the customers are acquired to join the membership or sign a contract, companies have a guarantee of continuous future cash flow for a period of time. Thus, companies in contractual business settings are concerned most with the expected duration or time of the relationship with customers, mainly because the duration of the relationship is often a key driver of future customer profitability. The key difference here from a modeling perspective is that in all contractual situations, the firm does observe when a customer is no longer a customer since the regular cash flows end. Because of the unique characteristics of non-contractual and contractual business settings, researchers need to build different models to study the issues in customer acquisition in contractual versus non-contractual settings.

We will provide additional discussion in each section where the contractual nature of the situation drives the model selection decision.

3.1.1 Data for empirical examples

In this chapter we will be providing a description of the key modeling frameworks that attempt to answer each key research question raised at the beginning of the chapter. We will also be providing at least one empirical example at the end of each subsection which will show how sample data can be used to answer these key research questions. For all the empirical examples in this chapter we provide a dataset titled ‘Customer Acquisition.’ In this dataset you will find a representative sample of 500 prospects from a typical B2B firm. The data include the following variables which will be used in some combination for each of the subsequent analyses:

Variable	
<i>Customer</i>	<i>Customer number (from 1 to 500)</i>
<i>Acquisition</i>	<i>1 if the prospect was acquired, 0 otherwise</i>
<i>First_Purchase</i>	<i>Dollar value of the first purchase (0 if the customer was not acquired)</i>
<i>CLV</i>	<i>The predicted customer lifetime value score. It is 0 if the prospect was not acquired or has already churned from the firm (000s)</i>
<i>Duration</i>	<i>The time in days that the acquired prospect has been or was a customer, right-censored at 730 d</i>
<i>Censor</i>	<i>1 if the customer was still a customer at the end of the observation window, 0 otherwise</i>
<i>Acq_Expense</i>	<i>Dollars spent on marketing efforts to try and acquire that prospect</i>

<i>Acq_Expense_SQ</i>	<i>Square of dollars spent on marketing efforts to try and acquire that prospect</i>
<i>Industry</i>	<i>1 if the prospect is in the B2B industry, 0 otherwise</i>
<i>Revenue</i>	<i>Annual sales revenue of the prospect's firm (in millions of dollars)</i>
<i>Employees</i>	<i>Number of employees in the prospect's firm</i>
<i>Ret_Expense</i>	<i>Dollars spent on marketing efforts to try and retain that customer</i>
<i>Ret_Expense_SQ</i>	<i>Square of dollars spent on marketing efforts to try and retain that customer</i>
<i>Crossbuy</i>	<i>The number of categories the customer has purchased</i>
<i>Frequency</i>	<i>The number of times the customer purchased during the observation window</i>
<i>Frequency_SQ</i>	<i>The square of the number of times the customer purchased during the observation window</i>

These data will be used for each of the examples presented at the end of each of the sections. These examples will cover the topics of response probability, number of newly acquired customers, initial order quantity, duration/time, and firm's performance.

3.2 Response probability

The first issue in customer acquisition is to model the probability of prospects being acquired. Hansotia and Wang [5] argued that prospects' response likelihood varies based on their profiles and the promotional materials they received. The authors adopted a logistic regression to model prospects' probability of response and used prospect profile variables as predictors. In logistic regression, because we can only observe whether the prospects respond or not, a latent response variable y^* indicating unobserved utility is assumed. Thus, y is usually defined such that

$$\begin{aligned} y_i^* &= \beta x_i + \varepsilon_i \\ y_i &= \begin{cases} 1 & \text{if } y_i^* > 0 \\ 0 & \text{if } y_i^* \leq 0 \end{cases} \end{aligned} \quad (3.1)$$

where y_i = the acquisition of customer i (1 = acquired, 0 = not acquired) and x_i = a vector of covariates affecting the acquisition of customer i . The probability that the prospect responds is given by

$$\Pr(y = 1) = \Pr(y_i^* > 0) = \Pr(\varepsilon_i > -\beta x_i). \quad (3.2)$$

For a logistic regression, ε_i has a logistic distribution, with mean 0 and variance equal to $\pi^2/3$. The cumulative distribution function of ε_i is expressed as

$$F_\varepsilon(\beta x_i) = \frac{1}{1 + \exp(-\beta x_i)} \quad (3.3)$$

and

$$\Pr(y = 1) = 1 - \Pr(\varepsilon_i \leq -\beta x_i) = 1 - F_\varepsilon(-\beta x_i) = \frac{1}{1 + \exp(-\beta x_i)}. \quad (3.4)$$

Setting the estimated value of $\Pr(y = 1)$ to $\hat{p}$, when β is estimated by $\hat{\beta}$, we have

$$\hat{p} = \Pr(y = 1) = \frac{1}{1 + \exp(-\beta x_i)}. \quad (3.5)$$

Equation 3.5, the probability of response p , is estimated by the log-odds function as the well-known logistic regression model

$$\ln\left(\frac{\hat{p}}{1 - \hat{p}}\right) = \hat{\beta}X. \quad (3.6)$$

To present how to estimate the logit model, we adopt Franses and Paap's (2001) introduction of the MLE method for the logit model. A brief introduction of MLE is provided in Appendix appA. The likelihood function for the logit model is the product of the choice probabilities over the i individuals, that is

$$L(\beta) = \prod_{i=1}^N (\Lambda(X_i\beta))^{y_i} (1 - \Lambda(X_i\beta))^{1-y_i}, \quad (3.7)$$

where $\Lambda(X_i\beta)$ is the cumulative distribution function according to the standardized logistic distribution, and the log-likelihood is

$$l(\beta) = \sum_{i=1}^N y_i \log \Lambda(X_i\beta) + \sum_{i=1}^N (1 - y_i) \log (1 - \Lambda(X_i\beta)). \quad (3.8)$$

Due to fact that

$$\frac{\partial \Lambda(X_i\beta)}{\partial \beta} = \Lambda(X_i\beta)(1 - \Lambda(X_i\beta))X'_i, \quad (3.9)$$

the gradient is given by

$$G(\beta) = \frac{\partial l(\beta)}{\partial \beta} = - \sum_{i=1}^N (\Lambda(X_i\beta))X'_i + \sum_{i=1}^N X'_i y_i, \quad (3.10)$$

and the Hessian matrix is given by

$$H(\beta) = \frac{\partial^2 l(\beta)}{\partial \beta \partial \beta'} = - \sum_{i=1}^N \Lambda(X_i\beta)(1 - \Lambda(X_i\beta))X'_i X_i. \quad (3.11)$$

By using prospects' profile variables as explanatory variables in the logistic regression model, the authors estimated an equation to predict each prospect's probability of responding to the acquisition offer. The estimated model can then be used to understand prediction accuracy within the same dataset (i.e., in-sample prediction) or score prospects not in the dataset for prediction (i.e., out-of-sample prediction). Prospects then can be sorted based on the predicted probability for the purpose of selection and optimal resource allocation.

Wangenheim and Bayón [11] argued that customer satisfaction influenced the number of WOM referrals which had an impact on customer acquisition. They asserted that the reception of a WOM referral has an increased marginal effect on the likelihood of a prospect to purchase. And they managed to answer two interesting questions related to customer acquisition: first, whether prospects' purchase likelihood is a function of WOM referrals; and, second, whether the characteristics of the WOM referrals resource including source expertise and similarity influence the probability that received WOM induces a purchase behavior. Since the authors conducted empirical studies in the energy market where the number of competitors is relatively small, they used customers' switching behavior as the dependent variable in modeling customer acquisition. To answer the first question, they only included the variable indicating the reception of WOM referral as the independent variable and used a binary logistic regression model to examine the effect that the independent variable has on the dependent variable:

$$\text{prob}(y = 1) = \frac{\exp(\beta \cdot x)}{1 + \exp(\beta \cdot x)} \quad (3.12)$$

where y indicated the switching behavior, and the binary independent variable x indicated whether WOM referral had been received. Thus, the effect of WOM referral on switching was obtained by subtracting the switching probability of a customer who had not received WOM ($x = 0$) from that of a customer who had received WOM ($x = 1$). To answer the second question, the authors only included the sample of prospects who had received WOM referral in the modeling procedure and also used a logit model to estimate whether switching behavior is a function of WOM referral resource characteristics, source expertise and similarity. which were included as independent variables, conditioning on receiving WOM referral:

$$\text{prob}(y = 1|x = 1) = \frac{\exp(\alpha \cdot z)}{1 + \exp(\alpha \cdot z)} \quad (3.13)$$

where z indicated the two independent variables, source expertise and source similarity.

A binary logit model is not the only model that can be used for response probabilities. The type of model depends on the assumption of the distribution of the error term. Referring back to Equation 3.1, if the error term ε has a logistic distribution, the corresponding model is a logit model; if the error term has a standard normal distribution, the corresponding model is a probit model. In this case the

cumulative distribution function of ε for a probit model is

$$F(x_i\beta) = \Phi(x_i\beta) = \int_{-\infty}^{x_i\beta} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right) dz. \quad (3.14)$$

The probit model is also estimated by the MLE method. Following Franses and Paap (2001), the relevant likelihood function is given by

$$L(\beta) = \prod_{i=1}^N (\Phi(X_i\beta))^{y_i} (1 - \Phi(X_i\beta))^{1-y_i}, \quad (3.15)$$

and the corresponding log-likelihood function is

$$l(\beta) = \sum_{i=1}^N y_i \log \Phi(X_i\beta) + \sum_{i=1}^N (1 - y_i) \log(1 - \Phi(X_i\beta)). \quad (3.16)$$

Differentiating $l(\beta)$ with respect to β gives

$$G(\beta) = \frac{\partial l(\beta)}{\partial \beta} = - \sum_{i=1}^N \frac{y_i - \Phi(X_i\beta)}{\Phi(X_i\beta)(1 - \Phi(X_i\beta))} \phi(X_i\beta) X'_i, \quad (3.17)$$

and the Hessian matrix

$$H(\beta) = \frac{\partial^2 l(\beta)}{\partial \beta \partial \beta'} = \sum_{i=1}^N \frac{\phi(X_i\beta)^2}{\Phi(X_i\beta)(1 - \Phi(X_i\beta))} X'_i X_i. \quad (3.18)$$

Similar to the logit model, the probit model is often used to model binary response variables, especially in cases where there is a desire to estimate a two-stage model. The probit model, while more complicated to estimate due to its lack of a closed-form solution, is theoretically appealing in a two-stage modeling framework due to its standard normal distribution. This makes it useful when the second stage is a least squares regression with a normally distributed error term. For example, Reinartz *et al.* [7] linked customer acquisition and relationship duration together using a probit two-stage least squares model. These authors used a probit model to determine the selection or acquisition process as shown by Equation 3.1 and predicted each prospect's response probability. The estimated probability was then included in the duration model as an independent variable to account for the interaction between acquisition and duration.

As the prospects' response likelihood is not always behaviorally observed, attitudinal propensity scale is also used to measure prospects' response intention. One might argue that attitudinal propensity measures are not perfectly correlated with behaviors, but researchers sometimes still use attitudinal dependent variables considering that attitude is usually positively related with behavior. Lix *et al.* [1] used two dichotomous attitudinal dependent variables, one indicating whether joined the membership (attitudinal) and the other one indicating whether used direct marketing channels (behavioral). The authors also used two interval-scale dependent variables, indicating prospects' attitudinal propensity toward pro-environment and US

products. The authors included 150 independent variables containing prospects’ demographic and life-style information in the regression and log-linear analysis. A brief introduction to log-linear analysis is provided in Appendix B. As Hansotia and Wang (1995), they used the estimated model to score samples in a holdout dataset and measured the effectiveness of their estimated models.

Whether it is the case that a logit or probit framework is used to model response probability, the output of the model is quite useful for determining which customer the firm is likely to acquire. In addition, the results of the binary choice model can provide the drivers of customer acquisition which can be useful for managers to make decisions in future customer acquisition campaigns.

3.2.1 Empirical example: Response probability

One of the key questions we want to answer with regard to customer acquisition is whether we can determine which future prospects have the highest likelihood of adoption. To do this we first need to know which past prospects were acquired and which were not. In the dataset provided for this chapter we have a binary variable which identifies whether or not a prospect was acquired by the firm (and hence became a customer) and a set of drivers which are likely to help explain a customer’s decision to adopt. At the end of this example you should be able to identify the following:

- 1. The drivers of customer acquisition likelihood.
- 2. The parameter estimates from the response probability model.

A B2B firm wants to improve the acquisition rate of customers and reduce the acquisition spending on prospects by better understanding which prospects are most likely to adopt. A random sample of 500 prospects (some of whom became customers) was taken from a recent prospect list. The information we need for our model includes the following list of variables:

Dependent variable	
<i>Acquisition</i>	<i>1 if the prospect was acquired, 0 otherwise</i>
Independent variables	
<i>Acq_Expense</i>	<i>Dollars spent on marketing efforts to try and acquire that prospect</i>
<i>Acq_Expense_SQ</i>	<i>Square of dollars spent on marketing efforts to try and acquire that prospect</i>
<i>Industry</i>	<i>1 if the prospect is in the B2B industry, 0 otherwise</i>
<i>Revenue</i>	<i>Annual sales revenue of the prospect’s firm (in millions of dollars)</i>
<i>Employees</i>	<i>Number of employees in the prospect’s firm</i>

In this case, we have a binary dependent variable (*Acquisition*) which tells us whether the prospect did adopt (= 1) or did not adopt (= 0). We also have five independent variables we believe will be drivers of adoption. First, we have how

many dollars the firm spent on each prospect (*Acq_Expense*) and the squared value of that variable (*Acq_Expense_SQ*). We want to use both the linear and squared term since we expect that for each additional dollar spent on the acquisition effort for a given prospect, there will be a diminishing return to the value of that dollar. Second, since the focal firm of this example is a B2B firm, the other three variables are firmographic variables of the prospects. These include whether the prospect sells to B2B (= 1) or B2C (= 0) customers (*Industry*), how much (in millions) that the prospect firms makes in annual revenue (*Revenue*), and how many employees the prospect firm has (*Employees*).

First, we need to model the probability that a prospect will adopt. Since our dependent variable (*Acquisition*) is binary, we select a logistic regression using the modeling framework as described earlier in the chapter (see Equation 3.1). We could also select a probit model and in general achieve the same results. In this case the y variable is *Acquisition* and the x variables represent the five independent variables in our database. When we run the logistic regression we get the following result:

Variable	Estimate	Standard error	p -value
<i>Intercept</i>	-26.206	3.537	<0.0001
<i>Acq_Expense</i>	0.067	0.010	<0.0001
<i>Acq_Expense_SQ</i>	-0.00004	0.000008	<0.0001
<i>Industry</i>	0.033	0.389	0.9326
<i>Revenue</i>	0.032	0.012	0.0070
<i>Employees</i>	0.005	0.001	<0.0001

As we can see from the results, four of the five independent variables are significant at a p -value of 5% or better with only *Industry* being statistically non-significant. First, this means that acquisition expense has a positive, but diminishing effect ($Acq_Expense > 0$ and $Acq_Expense_SQ < 0$) on acquisition likelihood. Second, it suggests that a prospect who is B2B (vs. B2C) will not matter in terms of acquisition likelihood, all else being equal. Third, the higher the *Revenue* that the prospect has, the more likely the prospect will be acquired. And, finally, the more *Employees* the prospect has, the more likely the prospect will be acquired.

It is also important to understand exactly how changes in the drivers of acquisition likelihood are likely to lead to either increases or decreases in acquisition likelihood. To do this we need to determine the odds ratio for each of the parameter estimates. Since we are dealing with a logistic regression, this means that we are interested in the log-odds ratio. For example, for $Revenue = x$,

$$\text{Odds}(\text{Acquisition} | \text{Revenue} = x) = \exp(-26.206 + 0.032x),$$

and, for $Revenue = x + 1$,

$$\text{Odds}(\text{Acquisition} | \text{Revenue} = x + 1) = \exp(-26.206 + 0.032(x + 1)).$$

By dividing the second equation by the first we get

$$\frac{\text{Odds}(\text{Acquisition}|\text{Revenue} = x + 1) = \exp(-26.206 + 0.032(x + 1))}{\text{Odds}(\text{Acquisition}|\text{Revenue} = x) = \exp(-26.206 + 0.032x)}.$$

We then simplify the equation to get the following:

$$\frac{\text{Odds}(\text{Acquisition}|\text{Revenue} = x + 1)}{\text{Odds}(\text{Acquisition}|\text{Revenue} = x)} = \exp(0.032) = 1.033$$

When we compute the log-odds ratio for each of the statistically significant variables we get the following results for an increase in 1 unit of the independent variable.

Variable	Log-odds ratio
<i>Acq_Expense</i>	$\exp(0.06696 - 0.00008 * \text{Acq_Expense})$
<i>Revenue</i>	1.033
<i>Employees</i>	1.005

We gain the following insights from the log-odds ratios. With regard to *Acq_Expense*, we see that the odds ratio is dependent on the level of *Acq_Expense*. This is due to the fact that we include both the level and squared terms for *Acq_Expense*. For example, if we usually spend \$500 on a given prospect, by spending \$501 we should see an increase in the likelihood of acquisition by $\exp(0.06696 - 0.00008 * 500) = \exp(0.0296) = 1.027$. This means that by increasing our spending from \$500 to \$501, we should see an increase in acquisition likelihood by 2.7%. It is also important to note that this will vary depending on the initial level of *Acq_Expense*. With regard to *Revenue*, we see that for each increase in *Revenue* by \$1 million the acquisition likelihood should increase by 3.3%. Finally with regard to *Employees*, we see that for each increase in *Employees* by 1 person the acquisition likelihood should increase by 0.5%.

As a result we now know how changes in acquisition expense and changes in prospect characteristics are likely to either increase or decrease our likelihood of acquisition. This information can provide significant insights to managers who are charged with determining the optimal amount of resources to spend on acquisition.

3.2.2 How do you implement it?

To implement the logistic regression in this example we used the PROC Logistic feature in SAS. While we did use SAS to estimate the model, many other statistical packages are capable of estimating a logistic regression including (but not limited to) SPSS, MATLAB, and GAUSS.

3.3 Number of newly acquired customers and initial order quantity

Companies will always want to acquire as many customers as possible and acquire those customers who will bring large orders for profit maximization. Companies usually perform certain marketing activities to encourage newly acquire customers to buy more the first time and in subsequent purchases. The operation of shipping fees, such as free shipping for large orders or normal shipping fees charged for small orders, is one common way to influence the purchase behavior of potential customers. Lewis [2] examined the data from an online retailing company which experimented with multiple shipping fee schedules for 502 days, including large order size incentives and penalties. The author wanted to examine the effects shipping fees on order incidence, order size, shipping revenues, and customer acquisition simultaneously. Thus, Lewis developed a system of linear regressions to account for the possible correlation among the dependent variables. Among these regressions equations, the author modeled the number of newly acquired customers and the average order size for new customers in the following two equations:

$$CA_{cq} = \beta_1 + \beta_2 x_1 + \beta_3 x_2 + \beta_4 x_3 + \beta_5 x_4 + \beta_6 x_5 + \varepsilon_1, \quad (3.19)$$

where CA_{cq} denotes the number of newly acquired customers, x_1 to x_5 denote the explanatory variables, including shipping, pricing, and customer base terms, and

$$Amount(new) = \beta_1 + \beta_2 x_1 + \beta_3 x_2 + \beta_4 x_3 + \beta_5 x_4 + \beta_6 x_5 + \varepsilon_2, \quad (3.20)$$

where $Amount(new)$ denotes the average order size for new customers, and x_1 to x_5 denote the shipping variables and pricing variable. And, to account for the possible correlation between the equations and the possible endogeneity of explanatory variables, the author used three-stage least squares in the estimation and used the lagged values of each quantity as instruments.

Besides shipping fees, Villanueva *et al.* [3] argued that marketing channels influence the customer acquisition process and the value that newly acquired customers will bring to the company. These authors developed two functions to measure how much new customers acquired through different acquisition channels, including WOM and marketing activities, contribute to firm performance. One function is called ‘the value-generating function,’ which links newly acquired customers’ contributions to the firm’s equity growth, and the other function is called ‘the acquisition response function,’ which expresses the interactions between marketing spending and the number of acquisitions. The authors argued that it was important to incorporate all the indirect effects of each acquisition on customer equity, not just each customer’s direct financial contribution to the company. As the estimated variables are treated as potentially

endogenous, the authors used a three-variable vector autoregression (VAR) modeling technique:

$$\begin{pmatrix} x_t \\ y_t \\ z_t \end{pmatrix} = \begin{pmatrix} a_{10} \\ a_{20} \\ a_{30} \end{pmatrix} + \sum_{l=1}^p \begin{pmatrix} a_{11}^l & a_{12}^l & a_{13}^l \\ a_{21}^l & a_{22}^l & a_{23}^l \\ a_{31}^l & a_{32}^l & a_{33}^l \end{pmatrix} \begin{pmatrix} x_{t-l} \\ y_{t-l} \\ z_{t-l} \end{pmatrix} + \begin{pmatrix} e_{1t} \\ e_{2t} \\ e_{3t} \end{pmatrix}, \quad (3.21)$$

where x denotes the number of customers acquired through the firm's marketing actions, y denotes the number of customers acquired through WOM, and z denotes the firm's performance. The subscript t denotes time, p denotes the lag order of the model, and $(e_{1t}, e_{2t}, e_{3t})'$ are white-noise disturbances distributed as $N(0, \Sigma)$. The direct effects are captured by a_{31}, a_{32} ; the feedback effects are captured by a_{13}, a_{23} ; the cross-effects are captured by a_{12}, a_{21} ; and the reinforcement effects are captured by a_{11}, a_{22}, a_{33} . The authors used impulse response functions (IRFs) to estimate the long-term effect of an unexpected shock in one variable on the other variables in the system. Dekimpe and Hanssens (2004) give a review of VAR models under the persistence modeling framework and we provide the related introduction in Appendix C.

Both of the previous examples we have discussed deal with cross-sectional time series data, or snapshots of cross-sections of customers that are repeatedly observed over time. There is a great benefit to having cross-sectional time series data – mainly because the effects that are uncovered by the modeling frameworks are persistent over time across customers. However, it might be the case that one of two limitations are present that do not enable you to use cross-sectional time series data. First, it is possible that the data you have is not repeatedly measured over time, that is, it is just cross-sectional. In this case you cannot use a time series modeling framework such as a VAR model. Second, it may be the case that there is no variation over time in any of your key drivers. For instance, if you always charge the same shipping costs over time, repeatedly measuring shipping costs in multiple time periods makes no sense. Thus, we also need to have a method available for instances when we only have cross-sectional data.

When only cross-sectional data are available it is rather straightforward to predict the number of newly acquired customers and the expected initial order quantity. For the number of newly acquired customers, all that we need to do is to determine the expected response probability of each prospect, as outlined in the previous subsection, and aggregate the prediction across all prospects. We can then compare the actual number of acquired customers to the predicted number of acquired customers.

For the initial order quantity we can run a two-stage regression, similar to the two-stage least squares regression as outlined in Reinartz *et al.* [7] where the acquisition probability is modeled using a probit model and the initial order quantity is modeled as a least squares regression that is conditional on the prospect being acquired. The result of this two-stage model is a prediction of expected initial order quantity conditional on acquisition probability that can be applied to any future prospect.

Thus, regardless of whether the data you have are cross-sectional or cross-sectional time series, this section describes various models that can be used for both number of newly acquired customers and initial order quantity.

3.3.1 Empirical example: Number of newly acquired customers

Besides being able to predict whether or not we are likely to acquire a prospect as we did in the previous example, we are also interested in determining how well our response probability model does in helping us accurately predict the total number of customers we are likely to acquire and specifically which prospects we are most likely to acquire. Thus, at the end of this example we should be able to do the following:

1. Predict the number of prospects we are likely to acquire.
2. Determine the accuracy of our prediction.

The information we need for this prediction includes the following list of variables:

Dependent variable

Acquisition 1 if the prospect was acquired, 0 otherwise

Independent variables

Acq_Expense Dollars spent on marketing efforts to try and acquire that prospect

Acq_Expense_SQ Square of dollars spent on marketing efforts to try and acquire that prospect

Industry 1 if the prospect is in the B2B industry, 0 otherwise

Revenue Annual sales revenue of the prospect's firm (in millions of dollars)

Employees Number of employees in the prospect's firm

In this case we only have cross-sectional data, that is, data from a single snapshot in time. Thus, we need to use the estimates we obtained from the response probability model in the first example to help us determine the predicted probability that each prospect will adopt. To do this we use the parameter estimates from the response probability model and values for the x variables to predict whether a customer is likely to be acquired. For a logistic regression we must apply the proper probability function as noted earlier in the chapter (see Equation 3.5):

$$P(\text{Acquisition} = 1|X\beta) = \frac{1}{1 + \exp(-X\beta)}.$$

For example, for Customer 9, when we input the statistically significant variables into the $X\beta$ computation we get $P_{[\text{Customer} = 9]}(\text{Acquisition} = 1|X\beta) = 0.06$ or 6%. Once we do this for each of the customers we can then decide which prospect we believe we are likely to acquire given our acquisition spending and on each prospect's characteristics.

Next, we need to create a cutoff value to determine at which point we are going to divide the prospects into the two groups – predicted to acquire and predicted not to acquire. There is no rule that explicitly tells us what that cutoff number should be. Often by default we select 0.5 since it is equidistant from 0 and 1. However, it is also reasonable to check multiple cutoff values and choose the one that provides the best predictive accuracy for the dataset. By using 0.5 as the cutoff for our example, any prospect whose predicted probability of acquisition is greater than or equal to 0.5 is classified as predicted to acquire, and the rest are predicted as not to acquire. To determine the predictive accuracy we compare the predicted to the actual acquisition values in a 2×2 table. For our sample of 500 we get Table 3.3.

As we can see from the table, our in-sample model accurately predicts 90.6% of the prospects who chose not to adopt (183/202) and 91.6% of the prospects who chose to adopt (273/298). This is a significant increase in the predictive capability of a random guess model¹ which would be only 58.4% accurate for this dataset. Since our model is significantly better than the best alternative, in this case a random guess model, we determine that the predictive accuracy of the model is good. If there are other benchmark models available for comparison, the ‘best’ model would be the one which provides the highest accuracy of both the prediction to acquire and not to acquire, or in other words the prediction would provide the highest sum of the diagonal. In this case the sum of the diagonal is 456 and it is accurate 91.2% of the time (456/500).

3.3.2 How do you implement it?

The implementation of this example was carried out using a SAS Data step and the Freq procedure. First we computed the predicted response probabilities for each prospect using the coefficients from the response probability example and scored the prospects as predicted to acquire (1) if the value was higher than the cutoff (0.5), and predicted not to acquire (0) if the value was lower than the cutoff. Next we ran PROC Freq and created a 2×2 table which compared the actual and predicted acquisition values to determine how well our model predicted the number of newly acquired customers.

Table 3.3 Predicted versus actual acquisition.

		<u>Actual acquisition</u>		<i>Total</i>
		0	1	
Predicted acquisition	0	183	19	202
	1	25	273	298
<i>Total</i>		208	292	500

¹ A random guess model would do the following. First, it would determine which bucket acquire or not acquire has more prospects in it. In this case 292 prospects were acquired versus 208 who were not. Then it would predict that all prospects would be acquired and it would be accurate 58.4% of the time (292/500).

3.3.3 Empirical example: Initial order quantity

Many firms have realized that it is not sufficient to merely focus on just trying to acquire as many customers as possible without any concern for the value that the customer is likely to provide. Research in marketing has shown that the initial order value can be a valuable predictor in a customer's future value to the firm – or at the least justify the amount of money that is spent on customer acquisition. Thus, it can be useful to understand the drivers of initial order value and in turn be able to predict each prospect's expected initial order value. At the end of this example we should be able to do the following:

1. Determine the drivers of initial order quantity (value).
2. Predict the expected initial order quantity for each prospect.
3. Determine the predictive accuracy of the model.

The information we need for this model includes the following list of variables:

Dependent variables

<i>Acquisition</i>	<i>1 if the prospect was acquired, 0 otherwise</i>
<i>First_Purchase</i>	<i>Dollar value of the first purchase (0 if the customer was not acquired)</i>

Independent variables

<i>Lambda(λ)</i>	<i>The computed inverse Mills ratio from the acquisition model</i>
<i>Acq_Expense</i>	<i>Dollars spent on marketing efforts to try and acquire that prospect</i>
<i>Acq_Expense_SQ</i>	<i>Square of dollars spent on marketing efforts to try and acquire that prospect</i>
<i>Industry</i>	<i>1 if the prospect is in the B2B industry, 0 otherwise</i>
<i>Revenue</i>	<i>Annual sales revenue of the prospect's firm (in millions of dollars)</i>
<i>Employees</i>	<i>Number of employees in the prospect's firm</i>

Again, we need to note that we are dealing with cross-sectional data. We see from the data requirement that in order to determine the drivers of initial order quantity we need to have two dependent variables: *First_Purchase* and *Acquisition*. This is due to the fact that expected initial order quantity is derived from the following equation:

$$E(\text{Initial Order Quantity}) = P(\text{Acquisition} = 1) \\ * E(\text{First_Purchase} | \text{Acquisition} = 1).$$

This equation shows us that the expected initial order quantity is a function of the probability that the prospect will be acquired multiplied by the expected value of a purchase given that the prospect was acquired. If we were to merely run a regression with *First_Purchase* as the dependent variable and ignore the probability that the prospect will be acquired, we would get biased estimates due to a potential sample selection bias.

Sample selection bias is a problem that is common in many marketing problems and has to be statistically accounted for in many modeling frameworks. In

this case the prospect has a choice of whether or not to be acquired before making a purchase. If we were to ignore this choice we would bias the estimates from the model and we would have fewer precise predictions for the value of *First_Purchase*. To account for this issue we need to be able to predict the value for both the probability of *Acquisition* (similar to what we have done for the first empirical example) and the expected value of *First_Purchase* given that the prospect is expected to be acquired. One important consideration to note is that we cannot just run two models independently since there is likely to be a correlation between the error terms of the two models. Thus, we need to use a modeling framework that can simultaneously estimate the coefficients of the two models, or at least account for the correlation between *First_Purchase* and *Acquisition*. To do this we use a two-stage modeling framework similar to that described earlier in this chapter and found in Reinartz *et al.* [7].

The first model for *Acquisition* will be set up using the same equation as for the response probability example (see Equation 3.1). The only difference here is that instead of using a logistic regression we will be using a probit model to estimate the coefficients. The main reason for this lies in the error term of the probit model which is distributed normal with a mean of 0 and a standard deviation of 1. The fact that the probit model and the OLS regression model (which we will be using for *First_Purchase*) are both normally distributed allows us to more easily estimate them in a two-stage framework.

Once we estimate the probit model we need to create a new variable, λ , which will represent the correlation in the error structure across the two equations. This variable, also known as the sample selection correction variable, will then be used as an independent variable in the *First_Purchase* model to remove the sample selection bias in the estimates. To compute λ we use the following equation, also known as the inverse Mills ratio:

$$\lambda = \frac{\phi(X'\beta)}{\Phi(X'\beta)}.$$

In this equation ϕ represents the normal probability density function, Φ represents the normal cumulative density function, X represents the value of the variables in the *Acquisition* model, and β represents the coefficients derived from the estimation of the *Acquisition* model.

Finally, we want to estimate a regression model for *First_Purchase* and include the variable ρ as an additional independent variable. This is done in a straightforward manner with the following equation:

$$\text{First_Purchase} = \gamma'\alpha + \mu\lambda + \varepsilon.$$

In this case *First_Purchase* is the value of the initial order quantity, γ is the matrix of variables used to help explain the value of *First_Purchase*, α are the coefficients for the independent variables, μ is the coefficient of the inverse Mills ratio, λ is the inverse Mills ratio, and ε is the error term.

When we estimate the two-stage model, we get the following parameter estimates for each of the two equations:

Acquisition model	Estimate	<i>First_Purchase</i> model	Estimate
<i>Intercept</i>	-15.134	<i>Intercept</i>	-62.261
<i>Acq_Expense</i>	0.039	<i>Acq_Expense</i>	0.705
<i>Acq_Expense_SQ</i>	-0.00002	<i>Acq_Expense_SQ</i>	-0.0008
<i>Industry</i>	0.051 ^a	<i>Industry</i>	-2.764 ^a
<i>Revenue</i>	0.018	<i>Revenue</i>	3.026
<i>Employees</i>	0.003	<i>Employees</i>	0.254
		<i>Lambda (λ)</i>	19.101

^aDenotes not significant at $p < 0.05$.

We gain the following insights from the results. We see that λ is positive and significant. We can interpret this to mean that there is a potential selection bias problem since the error term of our selection equation is correlated positively with the error term of our regression equation. We also see that all other variables of the *First_Purchase* model are significant with the exception of *Industry*, meaning that we have uncovered many of the drivers of initial order quantity. We find that *Acq_Expense* is positive with a diminishing return, as noted by the positive coefficient on *Acq_Expense* and the negative coefficient on *Acq_Expense_SQ*. We also find that two of the firm characteristic variables are positive (*Revenue* and *Employees*) showing that firms with higher annual revenue and firms with more employees tend to have larger initial order quantities.

Our next step is to predict the value of *First_Purchase* to see how well our model compares to the actual values. We do this by starting with the equation for expected initial order quantity at the beginning of this example:

$$E(\text{Initial Order Quantity}) - P(\text{Acquisition} = 1) * E(\text{First_Purchase} | \text{Acquisition} = 1) \\ = \Phi(X'\beta) * (\gamma'\alpha + \mu\lambda).$$

In this case Φ is the normal cumulative distribution function (CDF), X is the matrix of independent variable values from the *Acquisition* equation, β is the vector of parameter estimates from the *Acquisition* equation, γ is the matrix of independent variables from the *First_Purchase* equation, α is the vector of parameter estimates from the *First_Purchase* equation, μ is the parameter estimate for the inverse Mills ratio, and λ is the inverse Mills ratio. Once we have predicted the *First_Purchase* value for each of the prospects, we want to compare this to the actual value from the database. We do this by computing the mean absolute deviation (MAD) and mean absolute percent error (MAPE). The equations are as follows:

$$\text{MAD} = \text{Mean}\{\text{Absolute Value}[E(\text{Initial Order Quantity}) - \text{First_Purchase}]\} \\ \text{MAPE} = \text{Mean}\{\text{Absolute Value}[(E(\text{Initial Order Quantity}) \\ - \text{First_Purchase}) / \text{First_Purchase}]\}.$$

We find for the acquired customers that $MAD = 51.96$ and $MAPE = 18.69\%$. This means that on average each of our predictions of *First_Purchase* deviates from the actual value by \$51.96 or 18.69% (based on a $\text{Mean}(\text{First_Purchase}) = \372.47). If we were instead to use the mean value of *First_Purchase* (\$372.47) as our prediction for all prospects (this would be the benchmark model case), we would find that $MAD = 127.17$ and $MAPE = 135.48\%$. Thus, our model does a significantly better job of predicting the value of initial order quantity than the benchmark case.

3.3.4 How do you implement it?

In this example we used a two-stage least squares approach with a probit model for acquisition and a least squares regression for the initial order quantity. We used multiple procedures in SAS to implement this model. First we used PROC logistic with a probit link function to estimate the model of customer acquisition. Next we used a SAS Data step to compute the inverse Mills Ratio using the output of the probit model. Finally we ran an OLS regression using PROC Reg and added the inverse Mills ratio as an additional variable. While we did use SAS to implement this modeling framework, programs such as SPSS can be used as well.

3.4 Duration/time

In contractual settings, the time customers spent with a company is critical to the maintenance and development of a business. Lewis [4] argued that customer acquisition promotions influence the duration of customer–company relationships. In the newspaper subscription industry, the author used survival analysis to model the relationship between discount depth and the length of time as a subscriber. The author adopted accelerated failure time models (Kalbfleish and Prentice, 1980) estimated as shown in the following specification:

$$\log(t_i) = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \sigma \varepsilon_i, \quad (3.22)$$

where ε_i is a random disturbance term, x_i and x_i^2 are the discount and discount squared, and β_s and σ are parameters to be estimated. The author varied the restrictions on ε and σ and specified various forms of the hazard function. The author used the exponential model as a baseline model, which assumes that the error term ε has a standard extreme-value distribution and that σ is equal to 1. The author also used two generalized gamma models: one included only the discount as an independent variable, and the other included also the quadratic form of discount.

Kalbfleish and Prentice (1980) gave a brief introduction of the accelerated failure time model as below. Suppose that $Y = \log T$ is related to the covariate z via a linear model $Y = z\beta + W$, where W is an error variable with density f . Exponentiation gives $T = \exp(z\beta)T'$ where $T' = \exp(w) > 0$ has hazard function $\lambda_0(t')$, say, that is independent of β . It follows that the hazard function for T can be written in

terms of this baseline hazard $\lambda_0(\cdot)$ according to

$$\lambda(t; z) = \lambda_0(t \exp(-z\beta)) \exp(-z\beta). \quad (3.23)$$

The survival function is

$$F(t; z) = \exp \left[- \int_0^{t \exp(-z\beta)} \lambda_0(u) du \right] \quad (3.24)$$

and the density function is the product of Equations 3.15 and 3.16. To estimate the accelerated failure time model, the MLE method is usually used. Fanses and Paap (2001) introduced the estimation procedure of the accelerated failure time model for the Weibull distribution. We provide the reference in Appendix D.

Schweidel *et al.* [12] argued that time until acquisition, which is defined as the time that elapses before a prospective customer acquires a particular service, influences the duration that customers stay with the company. Using data from telecommunication services, the authors used a customer-level bivariate timing model for the time until acquisition of customers who eventually acquired service in the observation period. The authors assumed a parametric distribution for the probability of acquiring service at time t_A :

$$f(t_A|\Theta) = S_A(t_A - 1|\Theta) - S_A(t_A|\Theta) \quad (3.25)$$

where $S_A(t_A|\Theta)$ is the survival function of the parametric distribution, $t_A = 1, 2, \dots, T$, and T is the length of the observation period. In the empirical studies, the authors considered three sets of possible baseline hazard specifications for the acquisition processes: the Weibull, log-logistic and expo-power distribution. The hazard function and survival function of these three baseline distributions are listed in Table 3.4. We provide an introduction to survival analysis in Appendix G.

Table 3.4 Hazard function and survival function of three baseline distributions.

Baseline distribution	Hazard function	Survival function
Weibull	$h(t \gamma, \alpha) = \gamma\alpha t^{\alpha-1}$	$S(t \gamma, \alpha) = \exp(-\gamma t^\alpha)$
Log-logistic	$h(t \gamma, \alpha) = \frac{\gamma\alpha(\gamma t)^{\alpha-1}}{1+(\gamma t)^\alpha}$	$S(t \gamma, \alpha) = \frac{1}{1+(\gamma t)^\alpha}$
Expo-power	$h(t \gamma, \alpha, \theta) = \gamma\alpha t^{\alpha-1} \exp(\theta t^\alpha)$	$S(t \gamma, \alpha, \theta) = \exp(\gamma/\theta)^{(1-\exp(\theta t^\alpha))}$

Source: Schweidel, Fader, and Bradlow (2008).

3.4.1 Empirical example: Duration/time

Customer acquisition is of key importance to many firms. However, many firms also want to look beyond just the point of acquisition and answer questions such as, ‘How long will a newly acquired customer still be a customer?’ So in this example we will try to uncover the drivers of new customer duration. We expect that customer duration will be a function of the customer’s exchange characteristics and firmographic information. But, we also want to see if the acquisition effort by the firm affects each new customer’s duration with the firm. At the end of this example we should be able to:

- 1. Determine the drivers of new customer duration.
- 2. Predict the duration of each new customer.

The information we need for this model includes the following list of variables:

Dependent variables	
<i>Duration</i>	<i>The time in days that the acquired prospect has been or was a customer, right-censored at 730 d</i>
Independent variables	
<i>Acq_Expense</i>	<i>Dollars spent on marketing efforts to try and acquire that prospect</i>
<i>Acq_Expense_SQ</i>	<i>Square of dollars spent on marketing efforts to try and acquire that prospect</i>
<i>Ret_Expense</i>	<i>Dollars spent on marketing efforts to try and retain that customer</i>
<i>Ret_Expense_SQ</i>	<i>Square of dollars spent on marketing efforts to try and retain that customer</i>
<i>Crossbuy</i>	<i>The number of categories the customer has purchased</i>
<i>Frequency</i>	<i>The number of times the customer purchased during the observation window</i>
<i>Frequency_SQ</i>	<i>The square of the number of times the customer purchased during the observation window</i>
<i>Industry</i>	<i>1 if the prospect is in the B2B industry, 0 otherwise</i>
<i>Revenue</i>	<i>Annual sales revenue of the prospect’s firm (in millions of dollars)</i>
<i>Employees</i>	<i>Number of employees in the prospect’s firm</i>
<i>Censor</i>	<i>1 if the customer was still a customer at the end of the observation window, 0 otherwise</i>

In this case the sample data are from a B2B firm that actually observes when a customer churns, that is, a contractual setting. We see from the list of variables that we have one dependent variable (*Duration*) which is right-censored. Our observation window for the data is only two years (730 days) after the prospects were acquired. Thus, we only observe a customer leaving the firm if it happens before the end of the second year. As a result, the data for *Duration* is right-censored at 730 days – meaning that any customer whose *Duration* value is 730 in the data

table has yet to leave the firm at the end of the observation window. We also have 10 independent variables which we hope will explain the variation in each new customer's duration with the firm.

These independent variables include the money the firm spent on acquisition (*Acq_Expense* and *Acq_Expense_SQ*), the money the firm spent on customer retention (*Ret_Expense* and *Ret_Expense_SQ*), customer exchange characteristics (*Crossbuy*, *Frequency*, and *Frequency_SQ*), and firmographic variables (*Industry*, *Revenue*, and *Employees*).

To determine the drivers of customer duration we use the accelerated failure time model similar to Equation 3.22 earlier in this chapter. In this case we have the following equation:

$$\ln(\text{Duration}) = X'\beta + \sigma\varepsilon,$$

where $\ln(\text{Duration})$ is the natural logarithm of the time the customer was active with the firm, X is a matrix of the 10 independent variables, β is a vector of coefficients, σ is the scale parameter, and ε is the error term. We estimate the model by assuming that ε follows a Weibull distribution – common among accelerated failure time models.² We estimate the model only with the 292 prospects that were acquired as customers. We get the following results from the estimation:

Variable	Estimate	<i>p</i> -value
<i>Intercept</i>	2.837	<0.0001
<i>Acq_Expense</i>	0.007	<0.0001
<i>Acq_Expense_SQ</i>	−0.000 01	<0.0001
<i>Ret_Expense</i>	0.001	<0.0001
<i>Ret_Expense_SQ</i>	−0.000 000 04	0.017
<i>Crossbuy</i>	0.098	<0.0001
<i>Frequency</i>	0.111	<0.0001
<i>Frequency_SQ</i>	−0.001 ^a	0.173
<i>Industry</i>	0.524	<0.0001
<i>Revenue</i>	0.012	<0.0001
<i>Employees</i>	0.0001	<0.0001
<i>Scale</i>	0.138	
<i>Shape</i>	7.252	

^aDenotes not significant at $p < 0.05$.

We see that each of the variables in the model is significant to at least a level of 1% with the exception of *Frequency_SQ*. In addition we see that for both *Acq_Expense* and *Ret_Expense* there is a positive, but diminishing, effect on *Duration*. This suggests that the more the firm spends, up to a threshold, on both acquisition

² There are many other distributions which can be used including the exponential, gamma, normal, and log-normal among others. It is often worthwhile to try several distributions to see if the model fit and prediction improve.

and retention, the longer the customer is likely to stay with the firm. While we expect that spending on retention efforts is directly linked to the duration of a customer, we also see here that spending on acquisition also helps determine the duration of a customer. We see for *Crossbuy* that the more products the customer has purchased, the longer the expected *Duration*. We see a positive effect for *Frequency* with a negative, but not significant, squared term. This suggests that customers who do not purchase very often are more likely to churn than customers who purchase at a moderate to high frequency. We see that customers who are B2B have higher *Revenue*, and more *Employees* are more likely to have a longer *Duration*. Finally, we obtain both a *Scale* and a *Shape* parameter which will help us with the prediction of *Duration* for each of the customers.

Next, we want to see how well we are predicting *Duration* for the customers. Since we do not actually observe *Duration* for the customers who have yet to churn, we cannot validate our results on those customers. We can predict the time until expected churn for the customers who have yet to churn, but we can only test to see how well our results accurately predict *Duration* for the customers who have already churned (unless we used a holdout sample where we actually knew when the customers churned beyond two years).

So, in order to test the predictive accuracy of the model on the 157 non-censored customers who have already churned, we need to have an equation to predict the value of *Duration*. We start by referring back to the original equation we estimated:

$$\ln(\text{Duration}) = X'\beta + \sigma\varepsilon.$$

We just need to recognize that σ is the scale parameter (0.138) and ε is derived from the 50% percentile of the Weibull distribution, or $\ln(-\ln(1-p)) = \ln(-\ln(1-0.5)) = -0.367$. Then to compute *Duration* we just need take the inverse logarithm of the right hand side. We get the following:

$$\text{Duration} = \exp(X'\beta + 0.138^* - 0.367) = \exp(X'\beta - 0.051).$$

Now we need to compare the actual *Duration* values for the 157 customers who churned during the observation window to the predicted values of *Duration*. We find a MAD of 45.97 days and a MAPE of 13.88%. We can compare this to the benchmark case where our prediction of *Duration* is the mean of the non-censored values, which is 333 days. We get a MAD of 170.77 days and a MAPE of 171.92%. We see that our model does significantly better than the benchmark model in predicting the *Duration* of customers in our database.

3.4.2 How do you implement it?

We implemented this model using the PROC Lifereg procedure in SAS where the dependent variable was right-censored. It is also possible to estimate this model and other accelerated failure time models using other programs such as STATA and R, among others.

3.5 Firm's performance (LTV, CLV, and CE)

Making customer acquisition selection decisions on the basis of the response probability, initial order quantity, and duration is not enough. Companies should select prospects to acquire based on their lifetime contribution, which can be termed as LTV, CLV, or CE. Hansotia and Wang [5] in the last scenario of their studies considered that prospects' LTV varied due to their profiles and the promotion packages they received. These authors modeled the present value of revenue (PVR) and estimated LTV by subtracting the present value of cost of goods and the estimated marketing and operation costs for that customer. A common issue that occurs in the estimation of CLV is that observations may be right-censored. Some of the customers may have already lapsed in the observation period while others may still use the product or service beyond the observation period. Estimating the lifetime value of the customers who are still active over the observation period by OLS regression may underestimate their lifetime contribution to the companies. Hansotia and Wang [5] used the Tobit model to estimate the PVR for customers. The Tobit model assumes that there is a latent variable y_i^* which linearly depends on x_i via a vector of parameters β . And there is a normally distributed error u_i to capture the random fluctuation of the relationship. For the right-censored situation in which censoring occurs above a value y_L , the Tobit model is

$$\begin{aligned} y_i &= \begin{cases} y_i^* & \text{if } y_i^* < y_L \\ y_L & \text{if } y_i^* \geq y_L \end{cases} \\ y^* &= \beta x_i + u_i, \quad u_i \sim N(0, \sigma^2). \end{aligned} \quad (3.26)$$

The authors did not restrict the latent variables having a normal distribution but chose from a set of distributions including the gamma, log-normal, logistic, and Weibull distributions. In their studies, the authors presented the likelihood function, which is the probability of observing the sample values, as

$$L = \prod_{i=1}^M \{\text{Prob}[PVR = \text{observed value}]\}^S \{\text{Prob}[PVR > \text{observed value}]\}^{1-S} \quad (3.27)$$

where $S = 1$ if observation i is uncensored, 0 otherwise. The authors estimated the Tobit model using the method of maximum likelihood by the Lifereg procedure in SAS. We provide the introduction and estimation of the type 1 Tobit model in Appendix E. For the other types of Tobit models, readers should refer to Amemiya (1985).

In non-contractual settings, customers can terminate the relationship with companies without any notice. Thus, it is not easy to identify whether the customers have lapsed or not and whether the observation is censored or not. Reinartz *et al.* [7] considered that the account is still active and the duration is right-censored if the expected time until the next purchase exceeds the recency since last purchase.

Using the data from a B2B high-tech manufacturer, Reinartz *et al.* [7] adopted a standard right-censored Tobit model to estimate the duration with independent variables including the estimated response probability. These authors further estimated customer profitability with a standard right-censored Tobit model using a set of exogenous variables and including the predicted duration and response probability. Similar to Reinartz *et al.* [7], Thomas *et al.* [8] developed an ARPRO (Acquisition, Retention, and PROfitability) model which uses a standard right-censored Tobit model to estimate and predict the total long-term customer profitability as a function of both the customer's predicted relationship duration and acquisition probability. The authors found that decreasing marketing spending for a B2B firm and catalog retailer and increasing spending to customers for a pharmaceutical firm would lead to increases in total customer profitability.

In an online grocery retailing industry, Lewis [4] separated customers' purchase activities in the observation period into three quarters, each quarter's activities expressed as RFM measures. The author used a Tobit model to estimate the three-quarter activity as a function of second-quarter RFM scores. The estimated model was then used to predict the fourth-quarter activity and the customer asset value was then calculated as a total of observed expenditures and the predicted fourth-quarter activity.

Blattberg and Deighton [9] proposed to find the optimal level of acquisition spending based on the maximization of first-year value. These authors solved this issue using a tool called decision calculus, an approach to decision making in which managers break down a complex problem into smaller and simpler problems which are then incorporated into a model to solve the original complex problem. To calculate the optimal level of acquisition spending that maximizes first-year value, the authors asked managers four questions to determine the acquisition expenditure per prospect \$ A , the acquisition rate obtained as a result of that expenditure a , the ceiling rate (the highest possible number of customers the company could reasonably acquire in a given time period), and the margin generated by a customer in a year \$ m . The authors then drew a curve of the actual acquisition probability from the equation

$$a = \text{ceiling rate} \times [1 - \exp(-k_1 \times \$A)] \quad (3.28)$$

where k is a constant that controls the steepness of the curve. Next, the authors calculated the net contribution from acquiring a prospect in the first year ($a\$m - \A) and drew the second curve showing how the average value of acquired customers in the first year varies with spending on acquisition. Combining the two curves, the authors found the optimal level of acquisition spending at the peak of the second curve where acquisition should stop.

The fundamental question encountered in managing customer acquisition is to model the probability of being acquired for customers. The logit or probit model is the most common method used since they are by nature used to model binary outcomes, such as being acquired or not. In addition, logit or probit models are used to examine the effect of different customer characteristics and purchase behavior on

the probability. In some circumstances, researchers may want to model several acquisition metrics together, such as the number of newly acquired customers and the initial order quantity, and systems of linear regression are able to complete the task. In addition, it is possible that the acquisition metrics, such as the number of customers acquired through promotion and through networking, and company profitability, are correlated and variant across time, and VAR is a suitable modeling method to account for the correlation and time effects. Furthermore, researchers may want to model the duration of newly acquired customers and determine the effects that various explanatory variables have on the time to defect. Since most of the duration data contain censored observations, OLS regression would provide biased estimates. A classic parametric method for survival data is the accelerated failure time model, which can be estimated by the method of maximum likelihood. The variation of the accelerated failure time model, depending on the assumption of its disturbance distribution, such as the Weibull, exponential, gamma, log-logistic and log-normal distribution, is often used in duration modeling. Customer acquisition leads to firm profitability, which is the metric researchers have a strong interest to model. CLV is often used to represent firm profitability achievement and capability, and researchers often have to encounter censored observations while modeling CLV. Tobit models are essentially suitable methods in the modeling and have gained extensive popularity. Besides stochastic methods, researchers have used deterministic methods, such as decision calculus, to determine the optimal level of acquisition spending for profit maximization.

3.5.1 Empirical example: Firm's performance

The final step of the analysis is to determine whether the customers that were acquired are profitable. We also want to know whether we can determine the drivers of the customer profitability to see if future acquisition efforts can help lead to acquire a larger number of profitable customers. At the end of this example we should be able to:

1. Determine the drivers of customer profitability.
2. Determine the predictive accuracy of the customer profitability model.

The information we will need for this model includes the following list of variables for the 292 prospects that became customers (*Acquisition* = 1):

Dependent variables

<i>Censor</i>	<i>1 if the customer was still a customer at the end of the observation window, 0 otherwise</i>
<i>CLV</i>	<i>The predicted customer lifetime value score. It is 0 if the prospect was not acquired or has already churned from the firm (000s)</i>

Independent variables

<i>Acq_Expense</i>	<i>Dollars spent on marketing efforts to try and acquire that prospect</i>
--------------------	--

50 STATISTICAL METHODS IN CUSTOMER RELATIONSHIP MANAGEMENT

<i>Lambda(λ)</i>	<i>The computed inverse Mills ratio from the acquisition model</i>
<i>Acq_Expense_SQ</i>	<i>Square of dollars spent on marketing efforts to try and acquire that prospect</i>
<i>Ret_Expense</i>	<i>Dollars spent on marketing efforts to try and retain that customer</i>
<i>Ret_Expense_SQ</i>	<i>Square of dollars spent on marketing efforts to try and retain that customer</i>
<i>First_Purchase</i>	<i>Dollar value of the first purchase (0 if the customer was not acquired)</i>
<i>Crossbuy</i>	<i>The number of categories the customer has purchased</i>
<i>Frequency</i>	<i>The number of times the customer purchased during the observation window</i>
<i>Frequency_SQ</i>	<i>The square of the number of times the customer purchased during the observation window</i>
<i>Industry</i>	<i>1 if the prospect is in the B2B industry, 0 otherwise</i>
<i>Revenue</i>	<i>Annual sales revenue of the prospect's firm (in millions of dollars)</i>
<i>Employees</i>	<i>Number of employees in the prospect's firm</i>

In this case we have two dependent variables. One dependent variable, called *Censor*, helps us to delineate between those customers who are still customers and those customers who have already left the firm. Our second dependent variable is called *CLV*, or customer lifetime value. For the sake of this example the *CLV* score (in thousands) for each of the acquired customers has been predicted for you. Since *CLV* is a forward-looking measure (i.e., it measures the expected future profitability of a customer), the only customers with a *CLV* score are those who are still customers with the firm. These are the customers who meet the following two criteria – they have a value of 1 for *Acquisition* and a value of 1 for *Censor*.

Similar to the example for initial purchase quantity, we need to take into account both the probability that the customer is still a customer (*Censor*) and the expected future profitability of the customer (*CLV*). To do this we get the following equation:

$$E(\text{Future Profitability}) = P(\text{Censor} = 1) * E(\text{CLV} | \text{Censor} = 1).$$

In this case we also recognize that we will have to account for potential sample selection bias and estimate the model in a two-stage framework. Again, similar to the modeling framework for initial order quantity, we first model *Censor* as a probit using all of the independent variables. Then, we compute the inverse Mills ratio for those customers who have *Censor* = 1. Finally, we run an OLS regression with *CLV* as the dependent variable only for the 135 customers who are still customers. We use all the independent variables listed and the inverse Mills ratio (λ). See the initial order quantity example for a more detailed description of the steps. We get the following results:

Censor model	Estimate	CLV model	Estimate
<i>Intercept</i>	–32.151	<i>Intercept</i>	–2.613
<i>Acq_Expense</i>	0.070 ^a	<i>Acq_Expense</i>	0.009

<i>Acq_Expense_SQ</i>	-0.000 1	<i>Acq_Expense_SQ</i>	-0.000 01
<i>Ret_Expense</i>	0.006	<i>Ret_Expense</i>	0.004
<i>Ret_Expense_SQ</i>	0.000 000 5 ^a	<i>Ret_Expense_SQ</i>	-0.000 000 7
<i>First_Purchase</i>	-0.002 ^a	<i>First_Purchase</i>	0.003
<i>Crossbuy</i>	0.758	<i>Crossbuy</i>	0.203
<i>Frequency</i>	0.852	<i>Frequency</i>	0.155
<i>Frequency_SQ</i>	-0.004 ^a	<i>Frequency_SQ</i>	-0.005
<i>Industry</i>	4.588	<i>Industry</i>	0.566
<i>Revenue</i>	0.072	<i>Revenue</i>	0.008
<i>Employees</i>	0.002 ^a	<i>Employees</i>	0.0002 ^a
		<i>Lambda (λ)</i>	0.174 ^a

^aDenotes not significant at $p < 0.05$.

We gain the following insights from the results. We see that λ is positive, but not significant. Thus we cannot interpret this to mean that there is not a selection problem, only that it is unlikely given that the error term of our selection equation is not correlated with the error term of our regression equation. We also see that with the exception of *Employees*, the remaining variables in the *CLV* model are all significant at $p < 0.05$. This means that we have uncovered many of the key drivers of *CLV* for this set of customers. We find that *Acq_Expense* and *Ret_Expense* are positive with a diminishing return, as noted by the positive coefficient on the level term and the negative coefficient on the squared term. This suggests that the more you spend on both acquisition and retention efforts, up to a threshold, the higher the customer's expected lifetime value. For *First_Purchase* we see that there is a positive coefficient suggesting that the higher the initial purchase amount, the higher the expected customer lifetime value. For *Crossbuy* we see that there is a positive coefficient suggesting that the more products a customer purchases, the higher the expected lifetime value. We find that *Frequency* is positive with a diminishing return, as noted by the positive coefficient on the level term and the negative coefficient on the squared term. This suggests that customers who purchase consistently at a moderate rate over their tenure are most likely to have the highest lifetime value. Then in terms of firmographic variables we find that customers who are B2B (*Industry*) and customers who have higher *Revenue* are more likely to have a higher lifetime value than customers who are not in the B2B *Industry* and have a lower *Revenue*.

Our next step is to predict *CLV* for each of the customers and see if our predictions are accurate. We do this by starting with the equation for expected future profitability at the beginning of this example:

$$E(\text{Future Profitability}) = P(\text{Censor} = 1) * E(\text{CLV} | \text{Censor} = 1) \\ = \Phi(X'\beta) * (\gamma'\alpha + \mu\lambda).$$

In this case Φ is the normal CDF, X is the matrix of independent variable values from the *Censor* equation, β is the vector of parameter estimates from the *Censor* equation, γ is the matrix of independent variables from the *CLV* equation, α is the

vector of parameter estimates from the *CLV* equation, μ is the parameter estimate for the inverse Mills ratio, and λ is the inverse Mills ratio. Once we have predicted the *CLV* value for each of the prospects we want to compare this to the actual value from the database.

We do the comparison for the *CLV* values of the 135 customers where *Censor* = 1, that is, the customers that are still active with the firm. We find for these data that the MAD is 0.29 (or \$290) and the MAPE is 4.5%. For our benchmark model we use the mean of 6.58 as our predicted value for each customer. In this case we get a MAD of 0.71 (or \$710) and a MAPE of 11.05%. We see that our *CLV* model provides a significantly better prediction of *CLV* than the benchmark model. This shows that identifying the drivers of *CLV* can significantly help a firm understand which customers are most likely to be profitable in the future.

3.5.2 How do you implement it?

Similar to the example for initial order quantity, we use a two-stage modeling framework. We first use a binary probit model using PROC Logistic and a probit link function to model the probability that a customer has already quit. Next, we use a SAS Data step to compute the inverse Mills ratio. Finally, we use PROC Reg in SAS to determine the drivers of *CLV*. Other statistical programs such as SPSS can be used to run these two-stage least squares models.

3.6 Chapter summary

The purpose of this chapter was to explore the current models for customer acquisition and provide some empirical examples as to how firms can apply this knowledge. Customer acquisition is the first key step in the CRM process. We have also shown that when firms engage in optimal prospect selection by understanding the drivers of customer acquisition and the ‘right’ amount of customer acquisition effort by understanding the relationship between marketing spending and customer value, the result can generate significant customer and firm profitability.

Customer acquisition – SAS code

```

/* Import Data - Library: statcrm */
proc import out=statcrm.customer_acquisition
    datafile="C:\_Your_Data_Location_\Statistics in CRM\Customer
Acquisition\Customer Acquisition.xls" dbms=excel replace;
    range="'Customer Acquisition Data$'"; getnames=yes; mixed=no;
scantext=yes; usedate=yes; scantime=yes;
run; quit;
/* Response Probability */
proc logistic data=statcrm.customer_acquisition descending;
model acquisition=acq_expense acq_expense_sq industry revenue
employees/rsquare;
output out=statcrm.response_probability_pred p=pred;
run; quit;
/* Number of Acquired Customers */
data statcrm.response_probability_pred1;
set statcrm.response_probability_pred;
if pred >= 0.5 then pred_acq=1; else pred_acq=0;
act_acq=acquisition;
run; quit;
proc freq data=statcrm.response_probability_pred1;
table pred_acq * act_acq; run; quit;
/* Initial Order Quantity */
proc logistic data=statcrm.customer_acquisition descending;
model acquisition=acq_expense acq_expense_sq industry revenue
employees/link=probit;
output out=statcrm.first_purchase_probit xbeta=xb_probit;
run; quit;
data statcrm.first_purchase_imr;
set statcrm.first_purchase_probit;
imr_acquisition=(pdf('Normal',xb_probit))/(probnorm(xb_probit));
run; quit;
proc reg data=statcrm.first_purchase_imr;
model first_purchase=acq_expense acq_expense_sq industry revenue
employees imr_acquisition;
output out=statcrm.first_purchase_imr1 p=xbeta;
where first_purchase > 0;
run; quit;
data statcrm.first_purchase1;
set statcrm.first_purchase_imr1;
pred_fp=probnorm(xb_probit)*(xbeta);
ad=abs(first_purchase-pred_fp);
ape=ad/first_purchase;
ad1=abs(first_purchase-372.47);
apel=ad1/first_purchase;
run; quit;
proc sql; select mean(ad) as mad, mean(ape) as mape,
mean(ad1) as random_mad, mean(apel) as mapel
from statcrm.first_purchase1; quit;
/* Duration - Time */
proc lifereg data=statcrm.customer_acquisition;

```

```

model duration*censor(1) = acq_expense acq_expense_sq ret_expense
ret_expense_sq crossbuy frequency frequency_sq industry revenue
employees;
where acquisition = 1;
output out = statcrm.duration xbeta = xbp = pred sres = resid;
run; quit;
data statcrm.duration1;
set statcrm.duration;
pred_duration = exp(xb+0.138*(log(-log(1-0.5)))));
ad = abs(duration - pred_duration);
ad1 = abs(duration - 333.3165);
run; quit;
proc sql; select mean(duration) from statcrm.duration1 where
acquisition = 1 and censor = 0; quit;
proc sql; select mean(ad) as mad, (mean(ad/duration)) as mape,
mean(ad1) as random_mad, (mean(ad1/duration)) as mapel
from statcrm.duration1 where acquisition = 1 and censor = 0; quit;
/* Firm Performance */
proc logistic data = statcrm.customer_acquisition descending;
model censor = acq_expense acq_expense_sq ret_expense ret_expense_sq
first_purchase crossbuy frequency frequency_sq industry revenue
employees/link = probit;
output out = statcrm.clv1a xbeta = xb_censor;
where acquisition = 1;
run; quit;
data statcrm.clv1b;
set statcrm.clv1a;
imr_censor = pdf('Normal',xb_censor)/probnorm(xb_censor);
run; quit;
proc reg data = statcrm.clv1b;
model clv = acq_expense acq_expense_sq ret_expense ret_expense_sq
first_purchase
crossbuy frequency frequency_sq industry revenue employees
imr_censor;
where clv > 0;
output out = statcrm.clv1cp = pred_clv;
run; quit;
data statcrm.clv1d;
set statcrm.clv1c;
ad = abs(pred_clv - clv);
ad1 = abs(pred_clv - 6.58);
run; quit;
proc sql; select mean(clv) from statcrm.clv1d where censor = 1; quit;
proc sql; select mean(ad) as mad, (mean(ad/clv)) as mape,
mean(ad1) as random_clv, (mean(ad1/clv)) as mapel
from statcrm.clv1d where acquisition = 1 and duration = 730; quit;

```

Customer acquisition – SAS output

The SAS System
The LOGISTIC Procedure
Model Information

Data Set	STATCRM.CUSTOMER_ACQUISITION	
Response Variable	Acquisition	Acquisition
Number of Response Levels	2	
Model	binary logit	
Optimization Technique	Fisher's scoring	

Number of Observations Read	500
Number of Observations Used	500

Response Profile

Ordered Value	Acquisition	Total Frequency
1	1	292
2	0	208

Probability modeled is Acquisition = 1.

Model Convergence Status
Convergence criterion (GCONV = 1E-8) satisfied.

Model Fit Statistics

Criterion	Intercept and Covariates	
	Intercept Only	
AIC	680.968	206.464
SC	685.183	231.751
-2 Log L	678.968	194.464

R-Square	0.6205	Max-rescaled	R-Square	0.8354
----------	--------	--------------	----------	--------

Testing Global Null Hypothesis: BETA = 0

Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	484.5043	5	<.0001
Score	307.1902	5	<.0001
Wald	90.1336	5	<.0001

The SAS System

The LOGISTIC Procedure
Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	Chi-Square	Pr > ChiSq
Intercept	1	-26.2062	3.5373	54.8855	<.0001
Acq_Expense	1	0.0672	0.0104	42.0956	<.0001
Acq_Expense_SQ	1	-0.00004	7.989E-6	26.8366	<.0001
Industry	1	0.0327	0.3868	0.0072	0.9326
Revenue	1	0.0319	0.0118	7.2657	0.0070
Employees	1	0.00496	0.000735	45.5539	<.0001

56 STATISTICAL METHODS IN CUSTOMER RELATIONSHIP MANAGEMENT

Effect	Odds Ratio Estimates		
	Point Estimate	95% Wald Confidence Limits	
Acq_Expense	1.070	1.048	1.091
Acq_Expense_SQ	1.000	1.000	1.000
Industry	1.033	0.484	2.205
Revenue	1.032	1.009	1.057
Employees	1.005	1.004	1.006

Association of Predicted Probabilities and Observed Responses

Percent Concordant	97.5	Somers' D	0.950
Percent Discordant	2.5	Gamma	0.950
Percent Tied	0.0	Tau-a	0.463
Pairs	60736	c	0.975

The SAS System

```

The FREQ Procedure
Table of pred_acq by act_acq
pred_acq  act_acq
Frequency,
Percent ,
Row Pct ,
Col Pct ,    0,    1, Total
ffffffff^ffffffff^ffffffff
    0,    183,    19,    202
      , 36.60,  3.80, 40.40
      , 90.59,  9.41,
      , 87.98,  6.51,
ffffffff^ffffffff^ffffffff^
    1,     25,   273,   298
      ,  5.00, 54.60, 59.60
      ,  8.39, 91.61,
      , 12.02, 93.49,
ffffffff^ffffffff^ffffffff^
Total    208    292    500
      41.60  58.40 100.00

```

The SAS System

```

The LOGISTIC Procedure
Model Information
Data Set          STATCRM.CUSTOMER_ACQUISITION
Response Variable Acquisition          Acquisition
Number of Response Levels 2
Model             binaryprobit
Optimization Technique Fisher's scoring

Number of Observations Read 500
Number of Observations Used 500

```

Response Profile		
Ordered Value	Acquisition	Total Frequency
1	1	292
2	0	208

Probability modeled is Acquisition = 1.

Model Convergence Status

Convergence criterion (GCONV = 1E-8) satisfied.

Model Fit Statistics

Criterion	Intercept Only	Intercept and Covariates
AIC	680.968	204.122
SC	685.183	229.410
-2 Log L	678.968	192.122

Testing Global Null Hypothesis: BETA = 0

Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	486.8459	5	<.0001
Score	307.1902	5	<.0001
Wald	107.5310	5	<.0001

The SAS System

The LOGISTIC Procedure
Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-15.1339	1.8623	66.0378	<.0001
Acq_Expense	1	0.0389	0.00543	51.3430	<.0001
Acq_Expense_SQ	1	-0.00002	4.16E-6	33.5373	<.0001
Industry	1	0.0513	0.2154	0.0568	0.8117
Revenue	1	0.0177	0.00661	7.1859	0.0073
Employees	1	0.00285	0.000403	50.0781	<.0001

Association of Predicted Probabilities and Observed Responses

Percent Concordant	97.5	Somers' D	0.950
Percent Discordant	2.5	Gamma	0.950
Percent Tied	0.0	Tau-a	0.463
Pairs	60736	c	0.975

The SAS System

The REG Procedure

Model: MODEL1

Dependent Variable: First_Purchase First_Purchase

Number of Observations Read 292

Number of Observations Used 292

58 STATISTICAL METHODS IN CUSTOMER RELATIONSHIP MANAGEMENT

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	6743424	1123904	1212.34	<.0001
Error	285	264211	927.05463		
Corrected Total	291	7007635			
Root MSE		30.44757	R-Square	0.9623	
Dependent Mean		372.46868	Adj R-Sq	0.9615	
Coeff Var		8.17453			

Parameter Estimates

Variable	Label	DF	Estimate	Error	t Value	Pr > t
Intercept	Intercept	1	-62.26094	48.96509	-1.27	0.2046
Acq_Expense	Acq_Expense	1	0.70484	0.13030	5.41	<.0001
Acq_Expense_SQ	Acq_Expense_SQ	1	-0.00076743	0.00009221	-8.32	<.0001
Industry	Industry	1	-2.76418	3.70605	-0.75	0.4564
Revenue	Revenue	1	3.02602	0.11146	27.15	<.0001
Employees	Employees	1	0.25406	0.00476	53.33	<.0001
imr_acquisition		1	19.10122	9.65693	1.98	0.0489

The SAS System

```

mad   mape random_mad mape1
-----
51.95896 0.186859 127.1712 1.354829

```

The SAS System

The LIFEREG Procedure

Model Information

Data Set	STATCRM.CUSTOMER_ACQUISITION
Dependent Variable	Log(Duration)
Censoring Variable	Censor
Censoring Value(s)	1
Number of Observations	292
Noncensored Values	157
Right Censored Values	135
Left Censored Values	0
Interval Censored Values	0
Name of Distribution	Weibull
Log Likelihood	51.102184898

```

Number of Observations Read    292
Number of Observations Used    292

```

Fit Statistics

```

-2 Log Likelihood      -102.204
AIC (smaller is better) -78.204
AICC (smaller is better) -77.086
BIC (smaller is better) -34.083

```

Algorithm converged.

Type III Analysis of Effects

Effect	Wald		
	DF	Chi-Square	Pr > ChiSq
Acq_Expense	1	52.0446	<.0001
Acq_Expense_SQ	1	223.6009	<.0001
Ret_Expense	1	179.5271	<.0001
Ret_Expense_SQ	1	3.6092	0.0575
Crossbuy	1	213.8306	<.0001
Frequency	1	113.1688	<.0001
Frequency_SQ	1	4.6705	0.0307
Industry	1	424.5219	<.0001
Revenue	1	214.4291	<.0001
Employees	1	10.4581	0.0012

The SAS System

The LIFEREG Procedure

Analysis of Maximum Likelihood Parameter Estimates

Parameter	DF	Estimate	Standard Error	95% Confidence Limits		Chi-Square	Pr > ChiSq
Intercept	1	2.8692	0.3053	2.2709	3.4675	88.34	<.0001
Acq_Expense	1	0.0068	0.0009	0.0049	0.0086	52.04	<.0001
Acq_Expense_SQ	1	-0.0000	0.0000	-0.0000	-0.0000	223.60	<.0001
Ret_Expense	1	0.0011	0.0001	0.0009	0.0012	179.53	<.0001
Ret_Expense_SQ	1	-0.0000	0.0000	-0.0000	0.0000	3.61	0.0575
Crossbuy	1	0.0992	0.0068	0.0859	0.1125	213.83	<.0001
Frequency	1	0.1198	0.0113	0.0977	0.1419	113.17	<.0001
Frequency_SQ	1	-0.0014	0.0006	-0.0027	-0.0001	4.67	0.0307
Industry	1	0.5378	0.0261	0.4866	0.5889	424.52	<.0001
Revenue	1	0.0121	0.0008	0.0105	0.0137	214.43	<.0001
Employees	1	0.0001	0.0000	0.0000	0.0001	10.46	0.0012
Scale	1	0.1387	0.0080	0.1239	0.1552		
Weibull Shape	1	7.2123	0.4154	6.4425	8.0741		

The SAS System

ffffff

327.3949

The SAS System

mad mape random_mad mapel

ff

45.42903 0.138056 165.937 1.723717

The SAS System

The LOGISTIC Procedure

Model Information

Data Set	STATCRM.CUSTOMER_ACQUISITION	
Response Variable	Censor	Censor
Number of Response Levels	2	
Model	binary probit	
Optimization Technique	Fisher's scoring	

Number of Observations Read 292

Number of Observations Used 292

60 STATISTICAL METHODS IN CUSTOMER RELATIONSHIP MANAGEMENT

Response Profile

Ordered	Total
Value	Censor Frequency
1	1 135
2	0 157

Probability modeled is Censor = 1.

Model Convergence Status
Convergence criterion (GCONV = 1E-8) satisfied.
Model Fit Statistics

Criterion	Intercept	
	Intercept Only	Intercept and Covariates
AIC	405.139	66.965
SC	408.816	111.086
-2 Log L	403.139	42.965

Testing Global Null Hypothesis: BETA = 0

Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	360.1737	11	<.0001
Score	183.0936	11	<.0001
Wald	28.7364	11	0.0025

The SAS System

The LOGISTIC Procedure
Analysis of Maximum Likelihood Estimates

Parameter	DF	Estimate	Standard Error	Chi-Square	Pr > ChiSq
Intercept	1	-32.1509	13.5467	5.6327	0.0176
Acq_Expense	1	0.0695	0.0420	2.7354	0.0981
Acq_Expense_SQ	1	-0.00009	0.000036	6.3828	0.0115
Ret_Expense	1	0.00597	0.00279	4.5733	0.0325
Ret_Expense_SQ	1	4.555E-7	9.229E-7	0.2436	0.6216
First_Purchase	1	-0.00170	0.00784	0.0472	0.8279
Crossbuy	1	0.7579	0.2159	12.3223	0.0004
Frequency	1	0.8515	0.2529	11.3406	0.0008
Frequency_SQ	1	-0.00353	0.0115	0.0948	0.7581
Industry	1	4.5876	1.0493	19.1152	<.0001
Revenue	1	0.0718	0.0298	5.8001	0.0160
Employees	1	0.00183	0.00208	0.7773	0.3780

Association of Predicted Probabilities and Observed Responses

Percent Concordant	99.7	Somers' D	0.993
Percent Discordant	0.3	Gamma	0.993
Percent Tied	0.0	Tau-a	0.496
Pairs	21195	c	0.997

The SAS System

The REG Procedure
Model: MODEL1
Dependent Variable: CLV CLV
Number of Observations Read 135

Number of Observations Used			135		
Analysis of Variance					
		Sum of	Mean		
Source	DF	Squares	Square	F Value	Pr > F
Model	12	106.94992	8.91249	60.14	<.0001
Error	122	18.07983	0.14820		
Corrected Total	134	125.02974			
	Root MSE	0.38496	R-Square	0.8554	
	Dependent Mean	6.57930	Adj R-Sq	0.8412	
	Coeff Var	5.85110			

Parameter Estimates					
		Parameter	Standard		
Variable	Label	Estimate	Error	t Value	Pr > t
Intercept	Intercept	-2.61302	0.93166	-2.80	0.0059
Acq_Expense	Acq_Expense	0.00870	0.00286	3.04	0.0029
Acq_Expense_SQ	Acq_Expense_SQ	-0.00001041	0.00000265	-3.93	0.0001
Ret_Expense	Ret_Expense	0.00356	0.00040352	8.82	<.0001
Ret_Expense_SQ	Ret_Expense_SQ	-6.76812E-7	1.117292E-7	-6.06	<.0001
First_Purchase	First_Purchase	0.00259	0.00108	2.40	0.0181
Crossbuy	Crossbuy	0.20338	0.02013	10.11	<.0001
Frequency	Frequency	0.15521	0.03415	4.55	<.0001
Frequency_SQ	Frequency_SQ	-0.00485	0.00181	-2.69	0.0082
Industry	Industry	0.56644	0.07845	7.22	<.0001
Revenue	Revenue	0.00817	0.00370	2.21	0.0289
Employees	Employees	0.00016932	0.00028115	0.60	0.5481
imr_censor		0.17380	0.13503	1.29	0.2005

The SAS System
ffffff
6.579302

The SAS System

mad mape random_clv mapel
ffffff
0.289925 0.045168 0.708706 110484

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